

Physics-Driven Transfer Learning Approach for OSNR Prediction in Hollow-Core Fiber-Based Data Center Interconnect Networks

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Abstract This paper proposes a physics-driven transfer learning approach for Hollow-Core Fiber (HCF) OSNR prediction. Utilizing a theoretical analytical model for pre-training, it achieves robust predictions with minimal experimental samples, eliminating 17 dB cliffs, reaching 0.388 dB RMSE and 0.6 dB 95% error bounds. ©2026 The Author(s)

Introduction

The rapid growth of AI and cloud computing services demands higher capacity and lower latency in data center interconnects (DCI). As standard single-mode fibers (SSMF) approach the nonlinear Shannon limit, hollow-core fibers (HCF)[1, 2], such as nested anti-resonant nodeless fibers (NANF), emerge as ideal alternatives. By confining light within an air core via anti-resonant interference, they enable near-light-speed propagation and ultra-low nonlinearity[3, 4]. However, this guiding mechanism makes HCF transmission highly sensitive to physical parameters. According to loss scaling laws, minute micro-structural perturbations—arising from manufacturing tolerances, mechanical extrusion, or environmental changes—are nonlinearly amplified, disrupting phase-matching conditions[3, 5, 6] and causing unpredictable wavelength-dependent loss (WDL) variations. Furthermore, HCF's ultra-low backscat-

tering degrades traditional monitoring methods[7], making accurate, real-time OSNR prediction highly challenging.

Existing evaluation paradigms struggle with these complex characteristics [8]. Current HCF OSNR prediction primarily relies on theoretical analytical models and data-driven machine learning. Analytical models are often “rigid,” providing averaged expectations that fail to capture random micro-distortions, leading to severe prediction cliffs under extreme anomalous link states. Conversely, pure data-driven models possess strong nonlinear fitting capabilities but remain highly susceptible to overfitting given the extreme scarcity of measured field data in early-stage HCF deployments [9–11].

To overcome the “rigid fitting” of analytical models and “sample constraints” of pure data models, this paper proposes a physics-driven transfer learning framework for HCF OSNR prediction, as illustrated in Fig. 1(c). The model establishes a

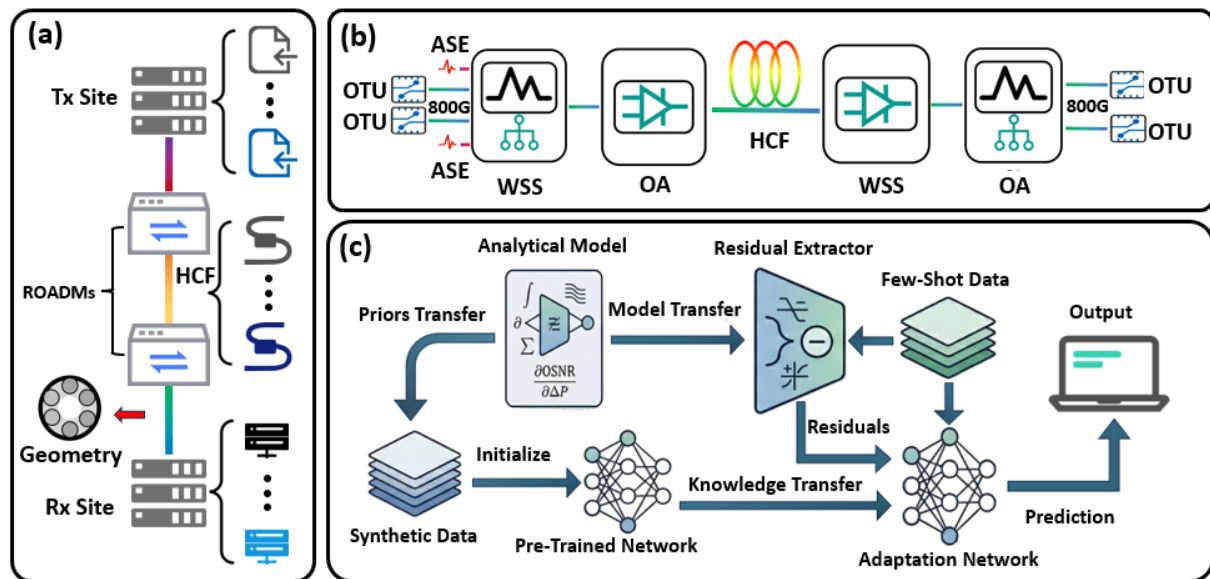


Fig. 1: System topology and the proposed prediction mechanism: (a) Physical topology of the cascaded ROADMs link for few-shot field data acquisition; (b) Hardware configuration of transceiver sites; (c) Architecture of the physics-driven transfer learning (PDTL) framework for OSNR prediction.

global cognitive foundation using theoretical analytical knowledge, followed by adaptive fine-tuning with a minimal amount of measured field data. Experimental results demonstrate that this framework effectively captures real-world network distortions, successfully eliminates prediction cliffs, and overcomes machine learning generalization bottlenecks at minimal data cost, achieving robust, high-precision HCF link evaluation.

Physics-driven Transfer Learning Mechanism

We propose the physics-driven transfer learning framework, as shown in Fig. 1(c), which overcomes the dual dilemmas of theoretical analytical models adhering to ideal assumptions and data-driven models risking overfitting in small-sample scenarios. By extracting fundamental evolution laws from a theoretical analysis model and performing residual space alignment, we transform hidden physical impairments into high-value neural network priors.

To inject strict physical consistency, the framework is grounded in the fundamental transmission evolution laws of HCF. HCFs, represented by NANF, rely on the cladding's F-P resonant cavity effect to guide light. Under ideal conditions, optical signal power evolution is affected by linear amplifier gains, nodal attenuations, and intrinsic fiber loss [3]:

$$P_{\text{out},k}(\lambda) = P_{\text{in},k}(\lambda) \cdot G_{\text{OA},k}(\lambda) \cdot A_{\text{WSS},k}(\lambda) \cdot \exp(-\alpha_{\text{HCF}}(\lambda)L_k) \quad (1)$$

The intrinsic attenuation coefficient, α_{HCF} , combines confinement loss ($\alpha_{\text{CL}} \propto \lambda^7 R^{-8}$) and surface scattering loss ($\alpha_{\text{SSL}} \propto \lambda^{-3} R^{-3}$). Due to these high-order nonlinear mapping relationships, minute variations in core geometry and operating wavelength are drastically amplified. Traditional theoretical models, constrained by rigid physical formulas, fail to capture the dynamic wavelength-dependent loss (WDL) distortion peaks of $\alpha_{\text{HCF}}(\lambda)$ caused by micro-structural deformations in live networks, leading to prediction cliffs of up to 17 dB.

To capture these ignored physical distortions, we perform rigorous full-link physical residual alignment. Considering the deterministic equipment interference, this paper calculates the physical power residual (ΔP_{dB}) in the logarithmic domain,

$$\Delta P_{\text{dB}} = P_{\text{launch,dBm}} + \sum G_{\text{OA,dB}} - \sum A_{\text{WSS,dB}} - P_{\text{rec,dBm}}. \quad (2)$$

In Eq. (2), ΔP_{dB} represents the deviation between the measured power P_{rec} and the ideal expectation derived from P_{launch} , G_{OA} , and A_{WSS} . This alignment is essential to eliminate "configuration bias," preventing the model from misin-

terpreting deterministic equipment settings as intrinsic fiber degradation. By decoupling these known factors, ΔP_{dB} acts as a physical feature filter to isolate hidden structural distortions (e.g., micro-structural-induced WDL), ensuring cross-configuration normalization and providing robust priors for fine-tuning under data scarcity.

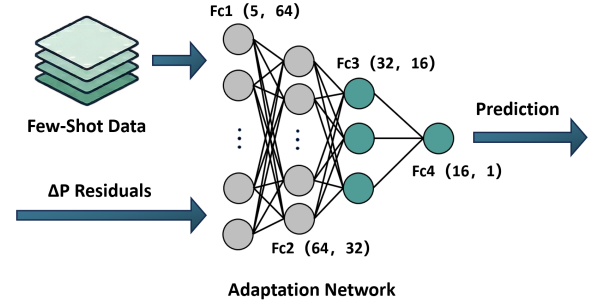


Fig. 2: Adaptation network: soft-frozen front-end (grey) and trainable high-order layers (colored).

To effectively compensate for these residuals, as illustrated in Fig. 2, we utilize a “pre-training/fine-tuning” differential transfer architecture. During pre-training, a four-layer multi-layer perceptron (MLP) is optimized on a large-scale dataset of 12,000 synthetic samples generated by the theoretical analysis model to allow the network to implicitly inherit HCF transmission mechanisms. During the live-network transfer stage (with 91 fine-tuning samples), a differential learning rate strategy is introduced. Specifically, an extremely small learning rate (5×10^{-6}) is applied to the front-end physical feature extraction layers to soft-freeze the learned physical laws, while a larger learning rate (5×10^{-4}) is assigned to the high-order output layers to focus on learning the live-network residuals. This dual-rate mechanism effectively prevents overfitting while ensuring the model's correction capability in complex HCF links.

Experimental Results and Analysis

The experimental evaluation compares the proposed PDTL mechanism with traditional analytical and purely data-driven (DD) models. The PDTL model is first pre-trained on 12,000 synthetic samples derived from physical scaling laws to establish a robust physical foundation. 107 real-world samples of operational HCF link are used to refine the prediction model, which are collected from China Telecom's live networks. Thereinto, 91 field samples enable model adaptation under data scarcity, while the remaining 16 field serve as an independent blind test set to verify prediction accuracy.

The OSNR scatter plots (Fig. 3) illustrate their fitting characteristics. The theoretical model (Fig. 3(a)) yields a rigid, averaged estimation, failing to capture dynamic fluctuations induced by physical perturbations. In contrast, both the data-driven model (Fig. 3(b)) and the TL-Model

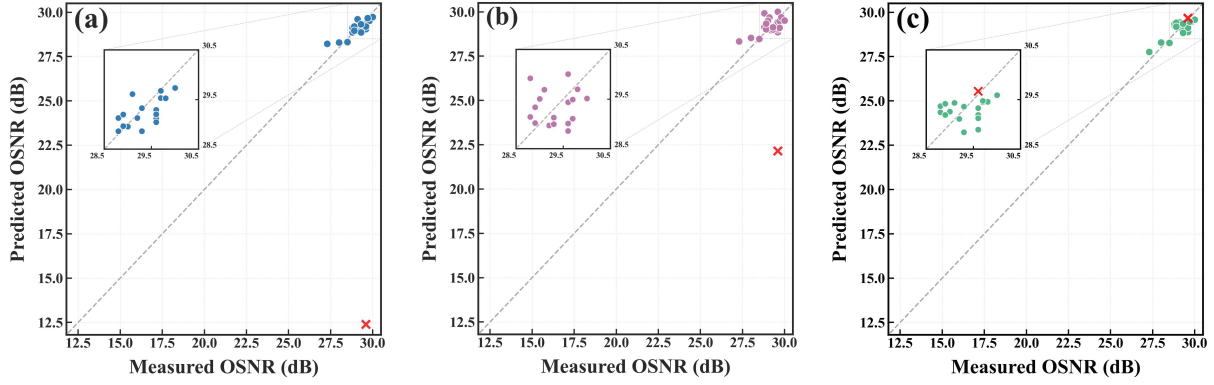


Fig. 3: Predicted versus measured OSNR distributions: (a) theoretical analytical model, (b) pure data-driven model, and (c) the proposed physics-driven transfer learning model.

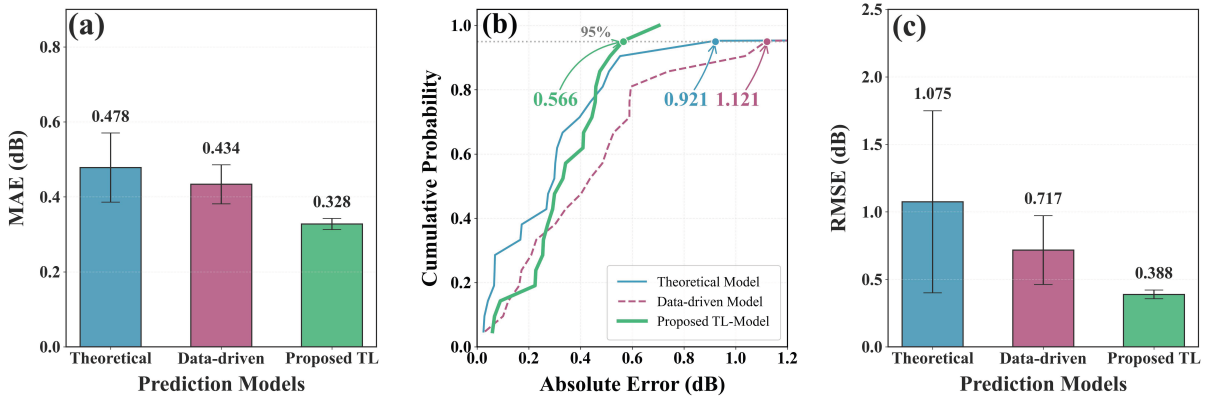


Fig. 4: Global statistical performance comparison for HCF OSNR prediction: (a) MAE comparison, (b) cumulative distribution function (CDF) of prediction errors, and (c) RMSE comparison.

(Fig. 3(c)) mitigate this systematic bias and adaptively track dynamic variations.

Beyond general tracking, practical HCF models must withstand extreme anomalous link states. Local structural distortions frequently trigger sudden signal attenuation [5]. Failing to account for these hidden impairments, the theoretical model (Fig. 3(a)) exhibits a severe prediction cliff, yielding an estimated OSNR of 12.5 dB against a measured value near 30 dB. This 17 dB deviation poses a direct risk of severe service interruptions in 800G networks. The data-driven model attempts to assimilate this distortion (Fig. 3(b)), elevating the outlier prediction to 22.5 dB, but it inevitably suffers from degraded generalization due to small-sample limitations. Conversely, the proposed TL-Model (Fig. 3(c)) successfully restores the extreme anomalous sample to the ideal $y = x$ diagonal. This confirms its capacity to reconstruct complex dynamic evolutions under data-constrained conditions.

These anomaly correction capabilities are corroborated by global statistical errors (Fig. 4). Due to idealized boundaries, the theoretical model yields high MAE (0.478 dB) and RMSE (1.075 dB). Constrained by the 91-sample dataset, the data-driven model suffers from overfitting [10], resulting

in 0.434 dB and 0.717 dB. In contrast, the TL-Model significantly reduces the MAE to 0.328 dB and the RMSE to 0.388 dB (a 64% reduction). Crucially, the CDF results in Fig. 4(b) show that the 95% confidence absolute error bound is compressed to 0.592 dB. This validates the framework's ability to eliminate generalization collapse and extreme prediction cliffs even under severe data constraints.

Conclusion

To address extreme parameter sensitivity and data scarcity in 800G live-network HCF links, this paper proposes a physics-driven transfer learning framework. By constructing a theoretical foundation with 12,000 simulated samples and applying a differential learning rate for fine-tuning on 91 field samples, the framework effectively captures complex micro-distortions. Results show the proposed PDTL eliminates 17 dB prediction cliffs, reducing MAE to 0.328 dB and RMSE to 0.388 dB. Crucially, the 95% confidence error bound is compressed within 0.6 dB, ensuring reliable link evaluation. This overcomes the generalization collapse of pure AI models under data constraints, providing a robust path for precise margin allocation and intelligent operation of future HCF networks.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (62271078).

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